rst_n

clk A[31:0]

B[31:0]

start

x_i__0

RTL_MUX

S=32'b00000000000000000000000000000000 I0

S=default I1 O

S[31:0]

x_i__1

RTL_MUX

S=32'b00000000000000000000000000000000 I0

S=default I1 O

S[31:0]-x0_i RTL_SUB

I0[31:0]

I1[31:0] O[31:0]

x_i

RTL_MUX

S=2'b00 I0[31:0] S=2'b01 I1[31:0] O[31:0] S[1:0]
x_i__2

RTL_MUX

S=2'b00 I0
S=2'b01 I1
S=2'b10 I2

S=default I3 O
S[1:0]

x_reg[31:0]

RTL_REG_ASYNC C

CECLR D Q >

x0_i__0 RTL_GT O

I0[31:0]
I1[31:0]

y_i__0

RTL_MUX

S=1'b1 I0

S=default I1 O S

y_i__1

RTL_MUX

S=32'b00000000000000000000000000000000 I0

S=default I1 O

S[31:0]

y_i__2

RTL_MUX

S=32'b00000000000000000000000000000000 I0

S=default I1 O

S[31:0]-y0_i RTL_SUB

I0[31:0]

I1[31:0] O[31:0]

y_i__3

RTL_MUX

S=2'b00 I0
S=2'b01 I1
S=2'b10 I2

S=default I3 O
S[1:0]

y_i

RTL_MUX

S=2'b00 I0[31:0] S=2'b01 I1[31:0] O[31:0] S[1:0]

y_reg[31:0]

RTL_REG_ASYNC C

CECLR D Q

st_i__0 RTL_ROM

A[31:0] O

st_i__1

RTL_MUX

S=32'b00000000000000000000000000000000 I0

S=default I1 O

S[31:0]

gcd_i__0 RTL_ROM

A[31:0] O

st_i__2

RTL_MUX

S=2'b00 I0
S=2'b01 I1
S=2'b10 I2

S=default I3 O
S[1:0]

st_i

RTL_MUX

V=B"01", S=2'b00 I0[1:0] V=B"10", S=2'b01 I1[1:0] S=2'b10 I2[1:0] S=default I3[1:0] O[1:0]

S[1:0]

gcd_i__1

RTL_MUX

S=32'b00000000000000000000000000000000 I0

S=default I1 O

S[31:0]

st_reg[1:0]

RTL_REG_ASYNC C

CECLR D Q

gcd_i

RTL_MUX

S=32'b00000000000000000000000000000000 I0[31:0] S=default I1[31:0] O[31:0] S[31:0] gcd_i__2
RTL_MUX

S=2'b00 I0
S=2'b01 I1
S=2'b10 I2

S=default I3 O
S[1:0]

busy_i__0

RTL_MUX

S=2'b00 I0
S=2'b01 I1
S=2'b10 I2

S=default I3 O

S[1:0] busy_i
RTL_MUX

S=2'b00 I0

S=2'b10 I1 O

S[1:0]

done_i

RTL_MUX

S=2'b00 I0
S=2'b01 I1
S=2'b10 I2

S=default I3 O
S[1:0]

gcd_reg[31:0]

RTL_REG_ASYNC C

CECLR D Q

busy_reg

RTL_REG_ASYNC C

CECLR D Q
done_reg

RTL_REG_ASYNC C

CLR D Q

gcd[31:0]

busy

done